SOFTWARE ENGINEERING LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Field	Value
Student Name	Akshat Sharma
SRN	PES1UG24CS042
Section	5A
Course	Software Engineering Lab
Problem Statement	#42 – Developer Tools & IT Operations
System	Internal Microservice Catalog & Health Portal
Jira Project	Health-Catalog (key: SELAB2)

Section 1: Agile Backlog Architecture (Epics & User Stories)

The functional requirements from Problem Statement #42 are converted into Agile Epics and User Stories. Each story is prioritized and estimated using Fibonacci-based story points.

Epic	Story ID	Actor	User Story	Priority	Story Points
Epic 1: Service Catalog & Documentation Management	FR-002	DevOps Engineer	As a DevOps Engineer, I want to register a new microservice by specifying its name, owner team, repository URL, API documentation endpoint, and health check URL, so that the service is added to the enterprise catalog and can be tracked.	High	3
Epic 1: Service Catalog & Documentation Management	FR-003	System Architect	As a System Architect, I want to visualize microservice dependencies as an interactive graph showing upstream and downstream connections, so that I can analyze the service topology and assess failure blast radius.	High	5
Epic 1: Service Catalog & Documentation Management	FR-005	Developer	As a Developer, I want to search and view aggregated OpenAPI/Swagger documentation for all registered microservices, so that I can discover and reuse existing APIs easily.	Medium	3
Epic 2: Health Monitoring & Alerting	FR-001	DevOps Engineer	As a DevOps Engineer, I want the system to automatically ping each registered microservice's health endpoint every 30 seconds and compute a rolling 24-hour availability percentage, so that I can monitor service reliability in real time.	High	5
Epic 2: Health Monitoring & Alerting	FR-004	DevOps Engineer	As a DevOps Engineer, I want to receive real-time downtime notifications via Slack and email when a service's health status changes to "Unhealthy", so	Medium	3

Epic	Story ID	Actor	User Story	Priority	Story Points
			that I can respond quickly and reduce mean time to detection.		

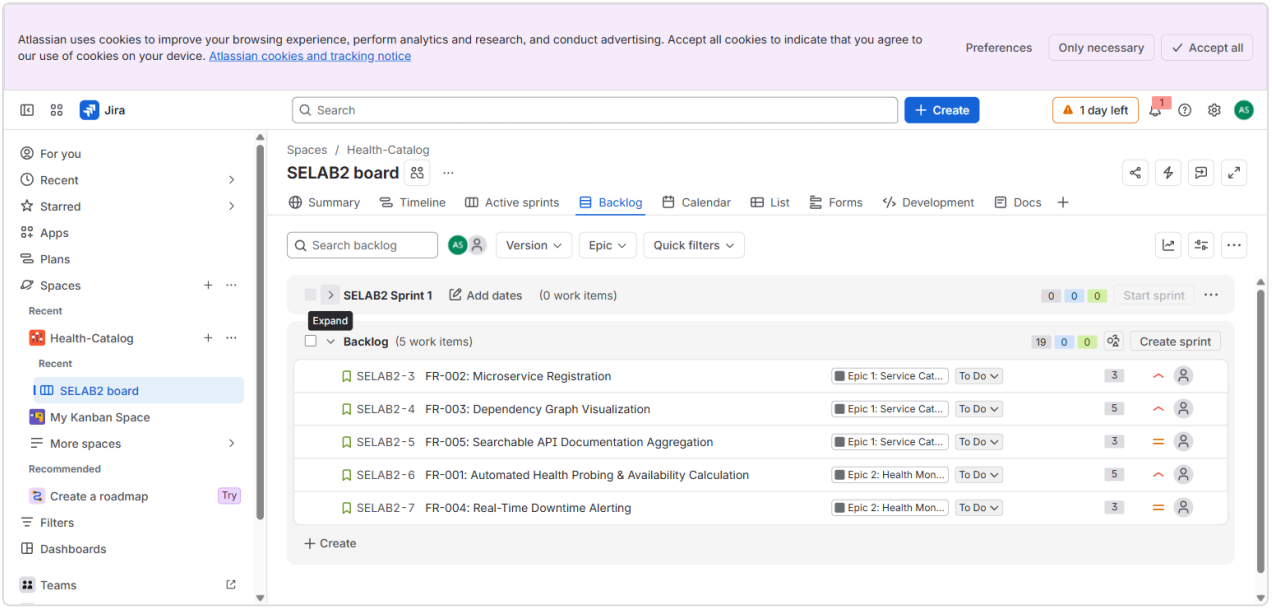
Total Backlog: 19 Story Points across 2 Epics and 5 User Stories.

Section 2: Sprint Execution & Board Simulation

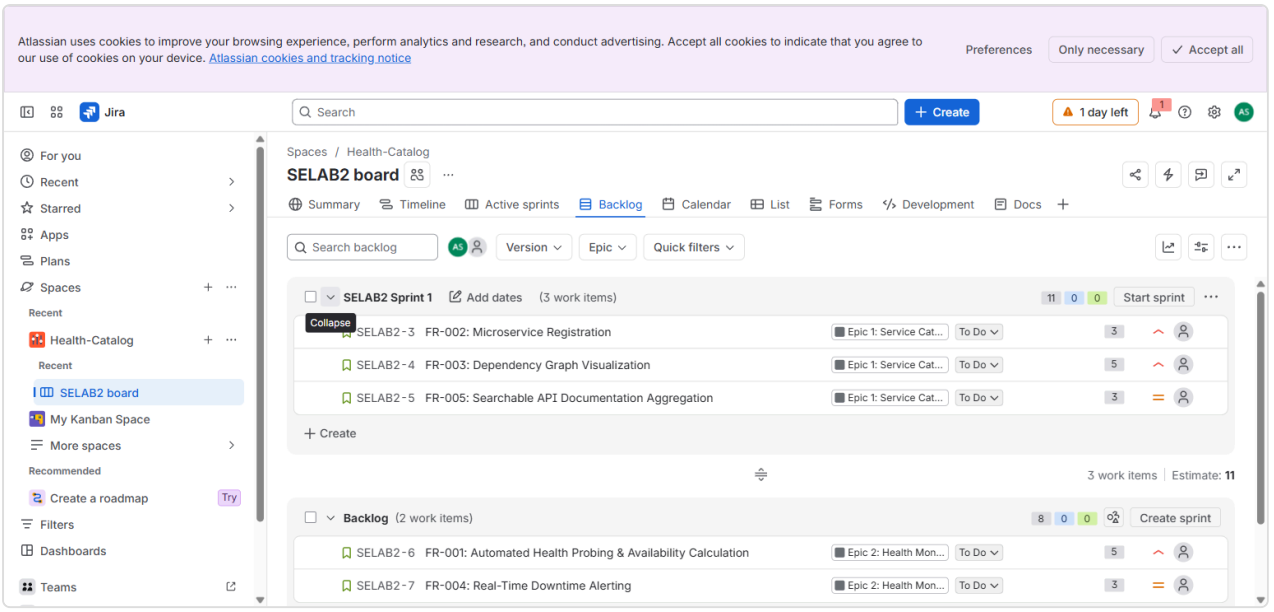
Sprint planning follows the Lab 2 requirement of a one-week sprint. The selected stories are moved through the Jira workflow: To Do → In Progress → Done.

Sprint	Duration	Committed Points	Stories	Sprint Goal
Sprint 1	1 week	11	FR-002, FR-003, FR-005	Establish the core service catalog by enabling microservice registration, dependency visualization, and API documentation aggregation.
Sprint 2	1 week	8	FR-001, FR-004	Deliver automated health monitoring and real-time downtime alerting so DevOps Engineers can detect and respond to outages quickly.

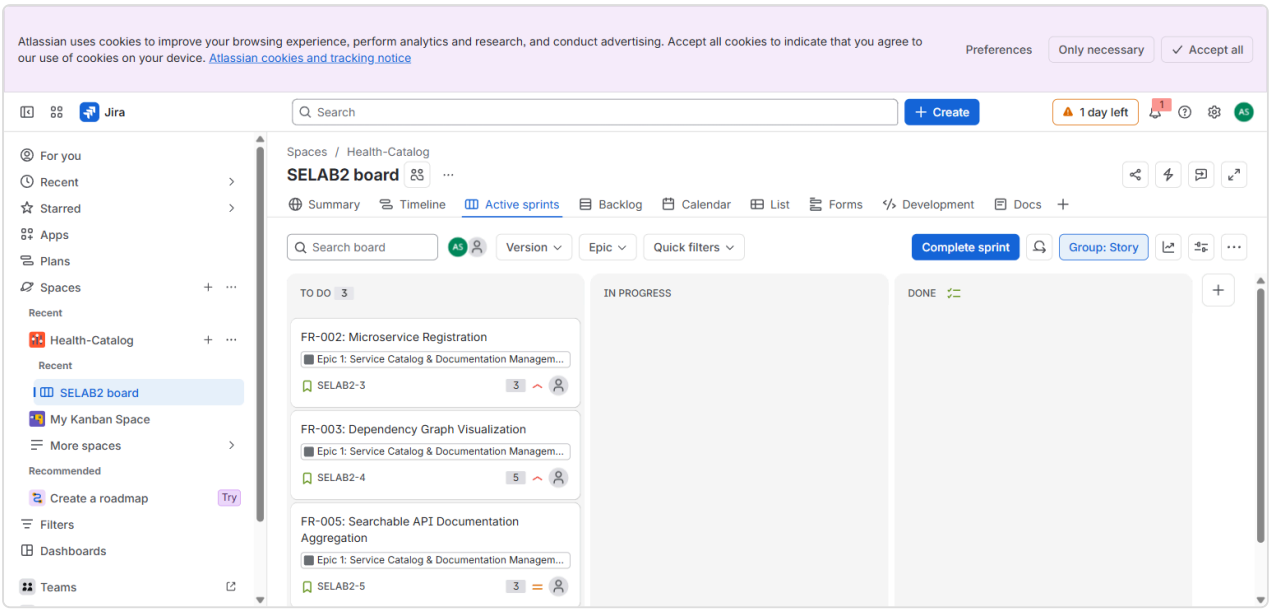
Screenshot 1 — Jira Backlog Evidence: full backlog with both Epics expanded and all 5 stories, story points and priorities visible.



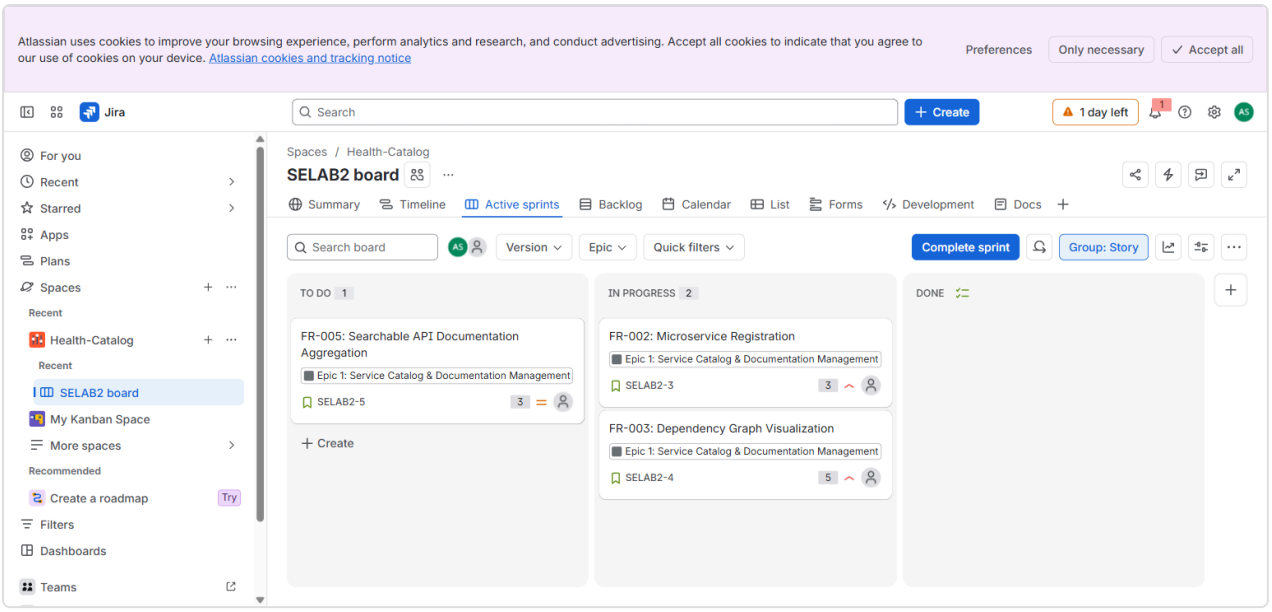
Screenshot 2 — Sprint 1 Planning: the 3 Sprint 1 stories (FR-002, FR-003, FR-005) added, right after clicking Start sprint.



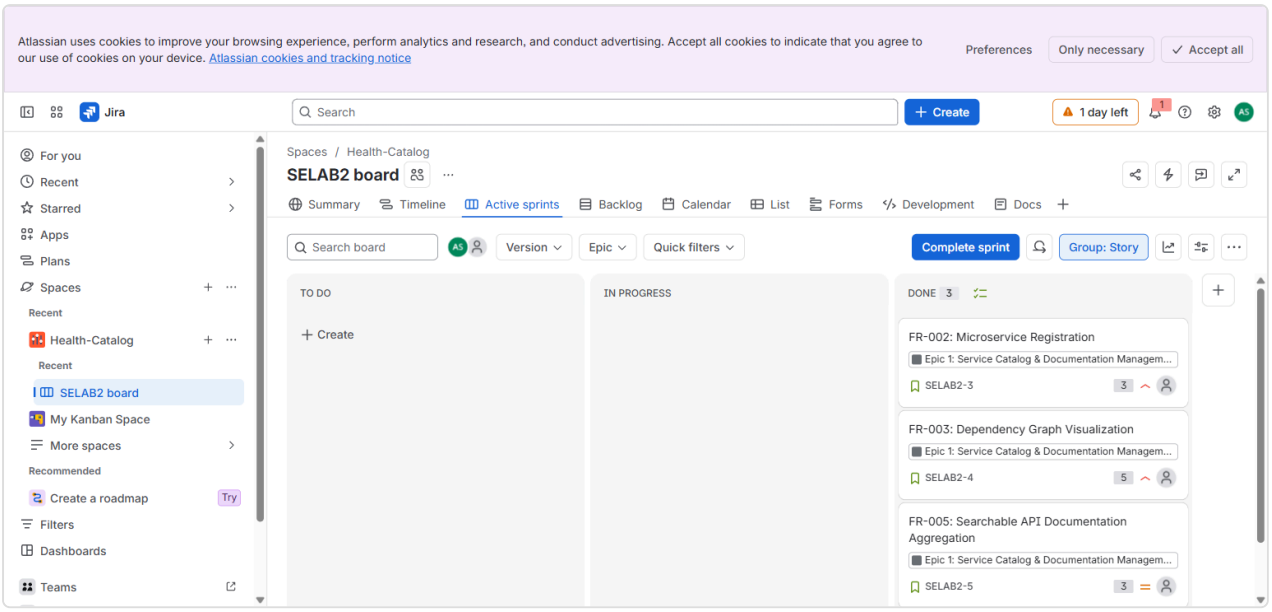
Screenshot 3 — Sprint 1 Active board (initial): all 3 cards in To Do immediately after the sprint starts.



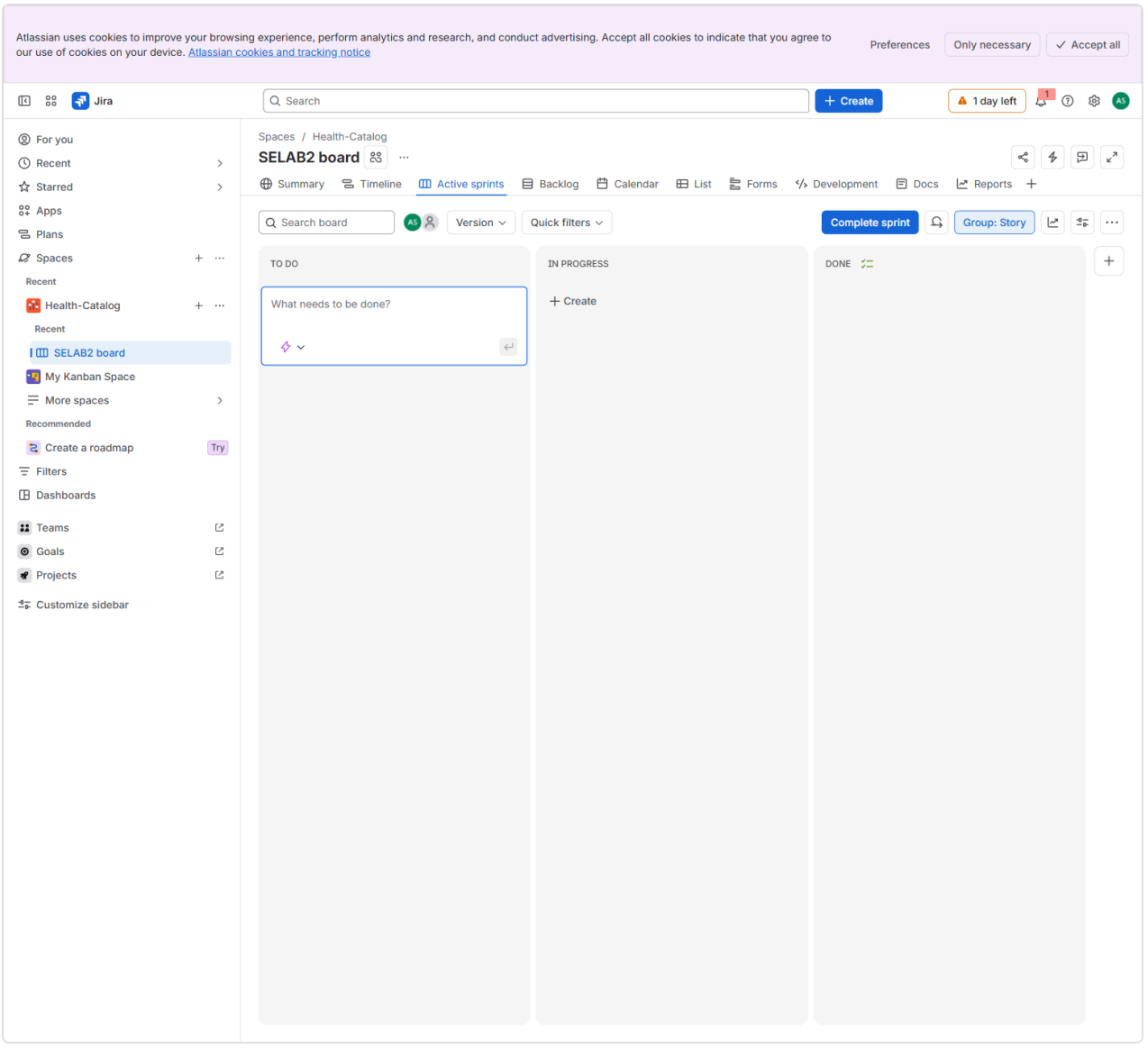
Screenshot 4 — Sprint 1 mid-execution: FR-002 and FR-003 moved to In Progress, FR-005 still in To Do.



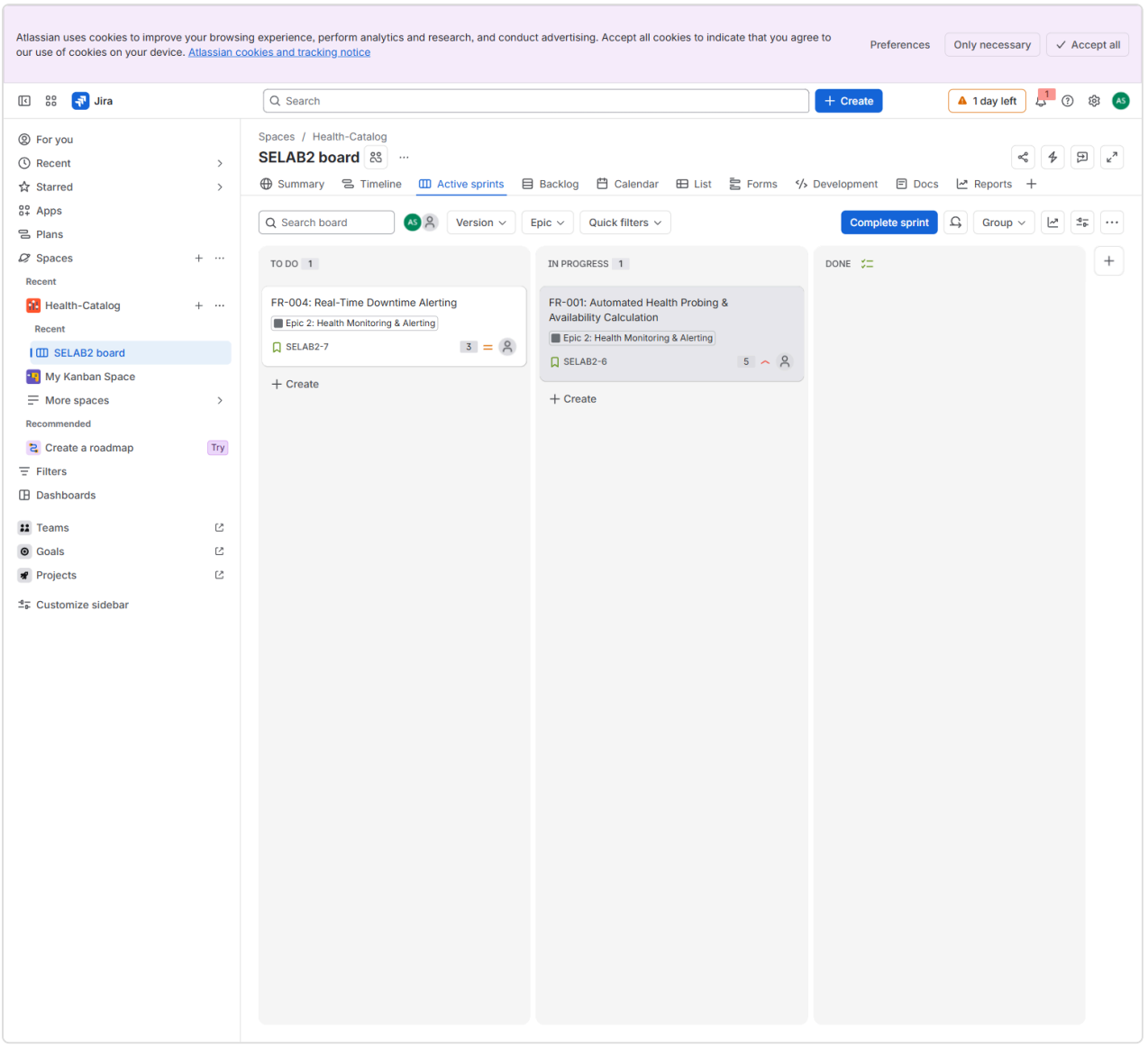
Screenshot 5 — Sprint 1 board just before completion: all 3 cards moved to Done.



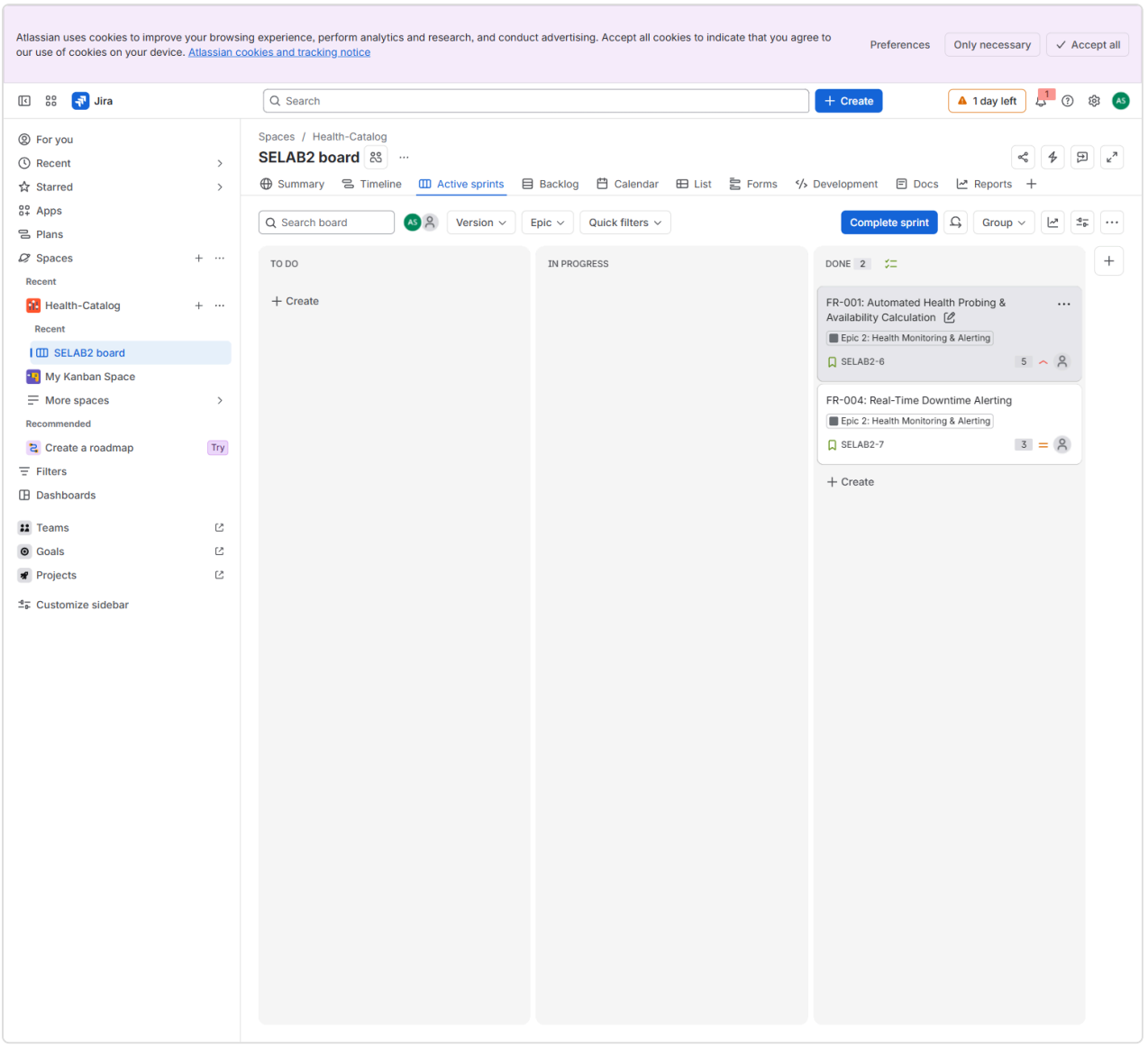
Screenshot 6 — Sprint 2 Active board (initial): both cards (FR-001, FR-004) in To Do right after Sprint 2 starts.



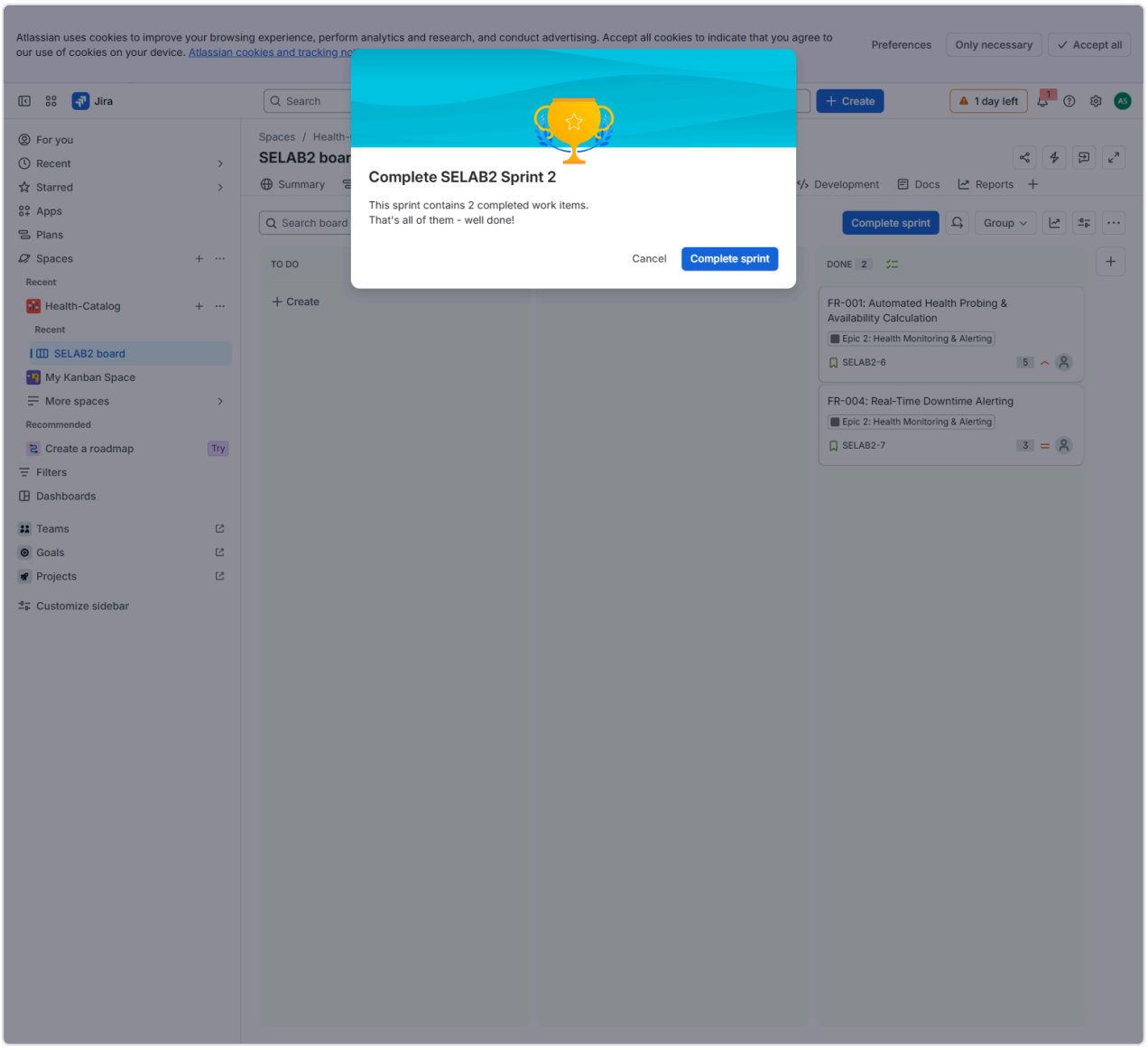
Screenshot 7 — Sprint 2 mid-execution: FR-001 moved to In Progress, FR-004 still in To Do.



Screenshot 8 — Sprint 2 board just before completion: both cards moved to Done.

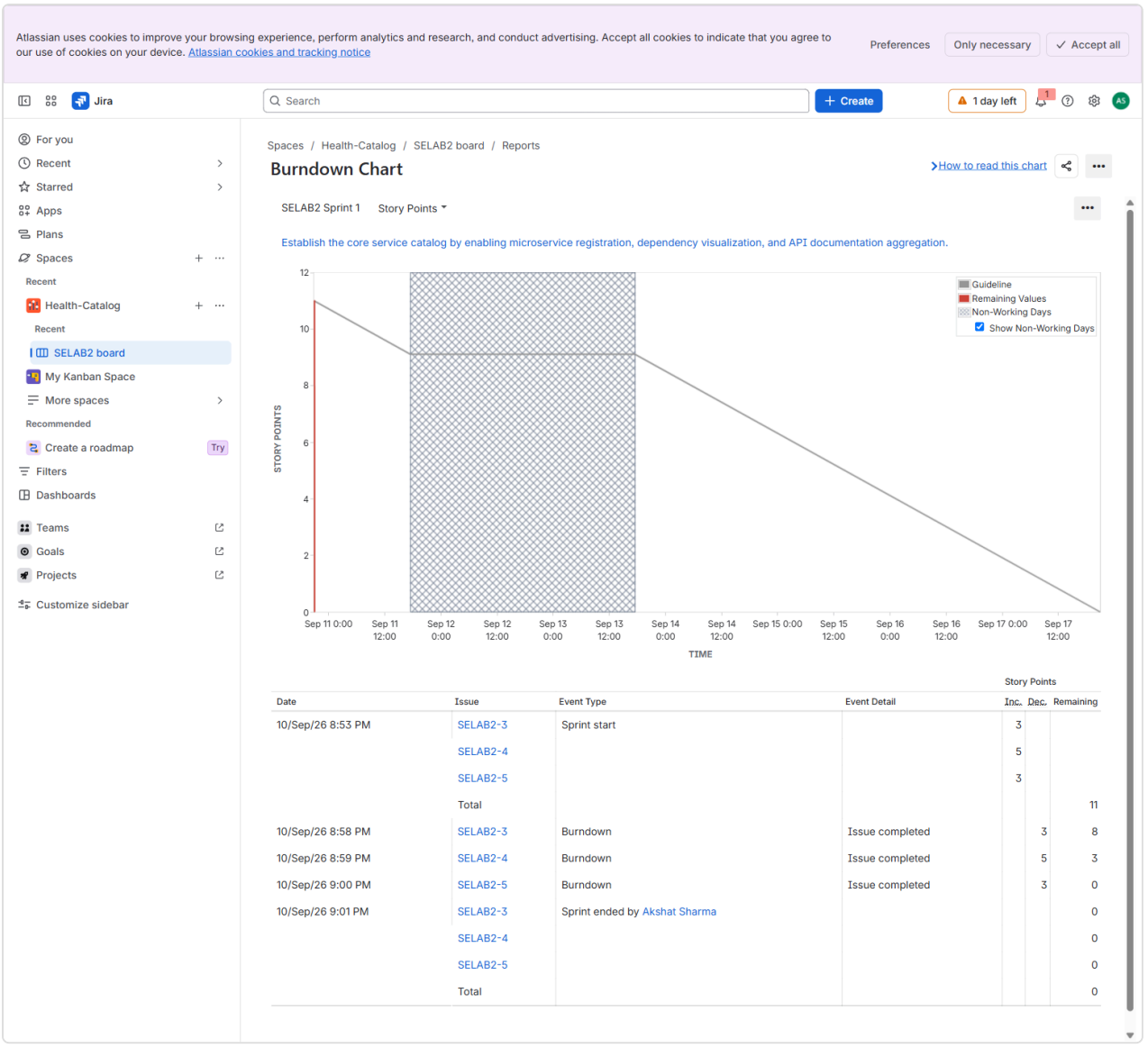


Screenshot 9 — Sprint 2 completion confirmation: "This sprint contains 2 completed work items. That's all of them - well done!"



Section 3: Sprint Burndown & Metrics Analysis

Screenshot 10 — Sprint 1 burndown chart:

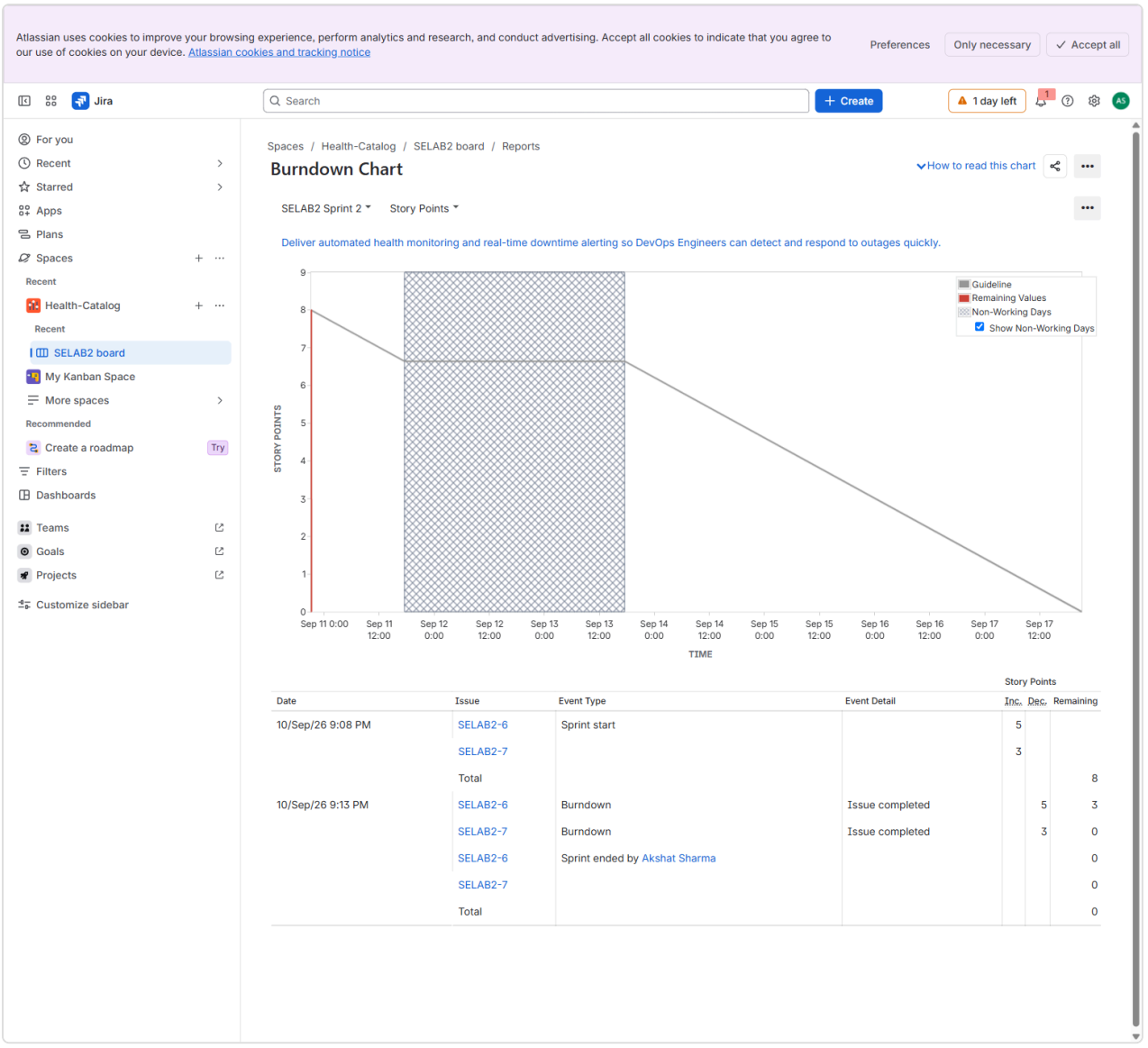


				Story Points		
Date	Issue	Event Type	Event Detail	Inc.	Dec.	Remaining
10/Sep/26 8:53 PM	SELAB2-3	Sprint start		3		
	SELAB2-4			5		
	SELAB2-5			3		
	Total					11
10/Sep/26 8:58 PM	SELAB2-3	Burndown	Issue completed		3	8
10/Sep/26 8:59 PM	SELAB2-4	Burndown	Issue completed		5	3
10/Sep/26 9:00 PM	SELAB2-5	Burndown	Issue completed		3	0
10/Sep/26 9:01 PM	SELAB2-3	Sprint ended by Akshat Sharma				0
	SELAB2-4					0
	SELAB2-5					0
	Total					0

Burndown Analysis — Sprint 1

Sprint 1 started with 11 committed story points across three work items: FR-002 (3 points), FR-003 (5 points), and FR-005 (3 points). The event log shows the sprint starting with all 11 points logged as remaining, followed by three "Issue completed" burndown events (one per story) that brought remaining work to 0 before "Sprint ended." Because the sprint was executed as a compressed, single-session simulation rather than over a real calendar week, the red remaining-work line drops as a steep vertical cliff immediately after the start, rather than the gradual day-by-day descent the grey guideline shows — i.e., burn-down happened far earlier than the ideal pace, since all committed work was finished almost immediately after sprint kickoff instead of being spread across the week.

Screenshot 11 — Sprint 2 burndown chart:



Burndown Analysis — Sprint 2

Sprint 2 started with 8 committed story points: FR-001 (5 points) and FR-004 (3 points). The event log records the sprint starting at 9:08 PM with 8 points remaining, both issues marked "Issue completed" by 9:13 PM (FR-001 first, dropping remaining to 3; then FR-004, dropping remaining to 0), and the sprint ended immediately after. As with Sprint 1, this produces a sharp early cliff-drop against the guideline rather than a steady multi-day burn, since the entire sprint was simulated within minutes rather than across its full 1-week window.

Section 4: Reflection & Analysis

Question 1: Did your estimations reflect the actual effort? The estimations were based on relative complexity: FR-001 and FR-003 were rated 5 points each because they involve non-trivial logic — rolling 24-hour availability computation with alert-triggering thresholds, and interactive dependency-graph rendering with pan/zoom and circular-dependency handling, respectively. FR-002, FR-004, and FR-005 were rated 3 points as more contained CRUD/notification/search features. Since both sprints were run as a compressed, single-session simulation rather than real elapsed development time, actual "effort" could not be measured in hours — but the relative point split held up conceptually: the two 5-point stories (FR-001, FR-003) were the two most functionally complex items in the backlog, and grouping them into separate sprints (FR-003 in Sprint 1, FR-001 in Sprint 2) kept each sprint's complexity balanced against its lower-point companions.

Question 2: Was your backlog well-prioritized? Yes — the High-priority items (FR-001, FR-002, FR-003) that form the foundation of the system (registering a service, seeing its dependencies, monitoring its health) were sequenced ahead of the Medium-priority items (FR-004, FR-005) that add value on top of that foundation (alerting, documentation search). This ordering makes sense because you cannot alert on a service's health (FR-004) before the service is registered and probed (FR-002, FR-001), and dependency visualization (FR-003) is more central to the "catalog" concept than searchable docs (FR-005).

Question 3: How did your simulated sprint align with your plan? Both sprints matched the plan exactly: Sprint 1 completed all 11 committed points (FR-002, FR-003, FR-005), and Sprint 2 completed all 8 committed points (FR-001, FR-004). Each story moved cleanly through To Do → In Progress → Done with no scope changes, no items carried over, and no items added mid-sprint — the "Complete sprint" confirmation for Sprint 2 explicitly confirmed "2 completed work items... that's all of them."

Question 4: What insights did the burndown chart give about your team's capacity? Both burndown charts show a steep, near-vertical drop to zero shortly after the sprint start rather than a gradual slope across the guideline — a direct result of simulating both 1-week sprints within a single session instead of across real elapsed days. In a real team setting this shape would be a red flag (either heavy under-scoping or work batched and reported all at once at sprint end); here it instead confirms that all 19 total story points across both sprints were fully deliverable with zero carryover, validating that the backlog was realistically scoped for a solo simulated team, even though the compressed timeline means the chart can't speak to actual day-to-day pacing or mid-week risk.